Symptomatic Bradycardia: Adult and Pediatric
C-2 Revised: June 10, 2024

Bradycardia is a heart rate less than 50 that may or may not cause symptoms of poor cardiac output. It may be seen as a normal variant in an acute myocardial infarction or as a result of a head injury or overdose.

In this protocol it is to be treated *ONLY* when the patient exhibits B/P < 90 systolic OR a decreased level of consciousness NOT associated with a suspected traumatic head injury, stroke, or other illness. Treatment for symptoms outside of these requires Medical Control orders.

In the event of beta blocker or calcium channel blocker overdose, refer to Suspected Overdose protocol.

Hypotension can be defined as a systolic blood pressure of less than the 5th percentile in a child/infant. Infant less than 1 month 60 mmHg systolic

1 month to 1 year	70 mmHg systolic
1 year to 10 years	70 plus 2 X age in years systolic
➢ 10 years	90 mm Hg systolic

Begin Initial Medical Care

If the bradycardic rhythm is Mobitz II or Third degree heart block especially with a QRS greater than 0.10 seconds, begin transcutaneous pacing at a rate of 80. If pacing does not improve cardiac output, or cannot obtain capture, administer atropine as outlined below.

Do not delay application of transcutaneous pacer if patient has a heart transplant as atropine will not work.

If a rhythm is other than those noted above,

- Administer **atropine 1.0 mg IV push rapidly initially.**
- Repeat **atropine 1.0 mg IV push** every 3-5 Minutes to a total of 3 mg.
- Do not delay application of transcutaneous pacing at a rate of 80 while awaiting IV access or for atropine to take effect if patient is severely symptomatic.

Adult Pain Management for Transcutaneous Pacing
INITIATE PACING FIRST. NEVER DELAY NEEDED PACING FOR PRE-EMPTIVE PAIN CONTROL/SEDATION.
Fentanyl 50 mcg slow IVP or IN (25mcg per nostril) for pain control if:
GCS ≥ 14
Pain scale > 5 (1-10 scale) 0r 4, 5 (Wong Baker Scale)
B/P ≥ 90 systolic
May be repeated every 10 minutes as long as the above parameters are still met.
Additional doses required before the 10 minute interval are AT MD DISCRETION ONLY.
OR
Versed: **10** mg midazolam IN (Intranasal) given as 5.0 mg (1.0 ml) in each nostril OR 2.5 mg to 5.0 mg slow IVP if:
GCS ≥ 14

- If the rate continues to be bradycardic ≤ 50 and hypotensive, B/P < 90 (and not hypovolemic), administer a 250 ml IV fluid bolus. If no change in B/P, initiate an epinephrine drip and administer at 2-10 mcg/min titrating for a HR > 60 and a B/P > 90 systolic.

Epinephrine MAY be prepared as 2 mg of a 1:1,000 solution in 500 ml of NS or D5W. This gives you a concentration of 4 mcg/ml.

1 mcg/min = 15 mcgtt
2 mcg/min= 30 mcgtt
3 mcg/min = 45 mcgtt
4 mcg/min = 60 mcgtt etc.

Epinephrine MAY be prepared as 4 mg of a 1:1,000 solution in 500 ml of NS or D5W if less fluid is desired. This gives you a concentration of 8 mcg/ml.

2 mcg/min = 15 mcgtt
4 mcg/min = 30 mcgtt
6 mcg/min = 45 mcgtt
8 mcg/min = 60 mcgtt

Bradycardia in infants and children is often managed by ensuring good oxygenation and ventilation.
Use length based tape to confirm dosing.

SYMPTOMATIC PEDIATRIC PATIENT ≤ 5 YEARS OLD: Patient is arousable, with delayed capillary refill, cool skin, weak peripheral pulses and a **HR < 80**

- Ensure oxygenation and ventilation
- Begin a fluid bolus of 20ml/Kg if no crackles present
- Administer epinephrine 1:10,000 0.01 mg/Kg or 0.1 ml/Kg IVP
- If no change may initiate transcutaneous pacing at a rate of 100.
- Epinephrine may be repeated every 3-5 minutes if no capture achieved, pacing does not increase cardiac output, or pacing cannot be tolerated.

SYMPTOMATIC PEDIATRIC PATIENT > 5 YEARS OLD: Patient is arousable, with delayed capillary refill, cool skin, weak peripheral pulses and a **HR < 60**

- Ensure oxygenation and ventilation
- Begin a fluid bolus of 20ml/Kg if no crackles present
- Administer epinephrine 1:10,000 0.01 mg/Kg or 0.1 ml/Kg IVP
- If no change may initiate transcutaneous pacing at a rate of 100.
- Epinephrine may be repeated every 3-5 minutes if no capture achieved, pacing does not increase cardiac output, or pacing is not tolerated.

UNSTABLE PEDIATRIC PATIENT: Patient is unconscious, **with** delayed capillary refill, cool skin, weak peripheral pulses, ***HR < 60 or B/P less than the 5th percentile.***

- Begin CPR and initiate transport
- Begin a fluid bolus of 20ml/Kg if no crackles present
- Administer epinephrine 1:10,000 0.01 mg/Kg or 0.1 ml/Kg IVP
- Initiate transcutaneous pacing at a rate of 100.
- Epinephrine may be repeated every 3-5 minutes if no capture achieved or pacing does not increase cardiac output Continue CPR until improvement in cardiac output.

Pain control for pacing is **"AT MD DISCRETION ONLY"**